SUMMER INTERNSHIP PROJECT

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I am truly thankful to Dr. Salkuti Surender Reddy, my internship supervisor, for his guidance, valuable suggestions, and constant encouragement during my work on Artificial Intelligence in the Operation of Microgrid. His knowledge in power systems and energy management helped me understand how advanced technologies can be applied in core electrical engineering. I am thankful to Woosong University for providing me the opportunity and resources to carry out this work. I also appreciate the support from my faculty coordinator and the Department of Electrical Engineering, NIT Kurukshetra, for their constant encouragement. Lastly, I acknowledge the role of IoT-based tools and technologies provided during the internship.

TABLE OF CONTENTS

S.NO	Title
1	Abstract
2	Introduction
3	About the organisation
4	Internship role and responsibilities
5	Learning and Observation
6	Project Report
7	Conclusion

ABSTRACT

This report presents my internship work on Artificial Intelligence in the Operation of Microgrid, carried out at Woosong University from May to July 2025. A microgrid is a small-scale power system that can operate independently or in connection with the main electricity grid, using renewable energy sources like solar and wind along with energy storage systems. The study focuses on how AI techniques, combined with Internet of Things (IoT) technology, can make microgrids smarter, more efficient, and reliable. AI applications such as energy forecasting, energy management, fault detection, grid optimization, and demand response were explored to improve system performance. IoT-based sensors and communication networks enabled real-time monitoring and control, helping in quick fault detection, efficient energy dispatch, and better load balancing. The project highlights how integrating AI and IoT can reduce costs, enhance reliability, and promote the use of clean energy, making microgrids an important part of future smart power systems.

INTRODUCTION

The main goal of my internship was to gain in-depth, practical experience in applying Artificial Intelligence (AI) and the Internet of Things (IoT) to enhance the efficiency, reliability, and automation of microgrid operations. Microgrids are a vital component of modern power systems, especially with the increasing integration of renewable energy sources. During my internship I learned how AI algorithms analyze this vast amount of data to identify patterns, optimize energy flow, predict potential faults, and support decision-making in real time. IoT devices played a critical role in enabling remote monitoring and seamless control, allowing for quick response to load variations, generation fluctuations, and system disturbances. This hands-on exposure allowed me to bridge the gap between the theoretical concepts I had studied in my electrical engineering coursework and their real-world application in modern, intelligent power distribution systems.

In the field of electrical engineering, internships are an essential stepping stone between academic learning and professional practice. They provide students with the opportunity to work on actual electrical infrastructure, understand the functioning of industry-standard equipment, and solve practical challenges that engineers face on a daily basis. I also gained insights into cutting-edge advancements like smart grids and demand-response systems. Beyond the technical skills, the internship helped me develop equally important soft skills such as effective communication, teamwork, adaptability, and project management. These are invaluable when collaborating with multidisciplinary teams and addressing the complex challenges of the modern energy sector. I carried out my internship at Woosong University, a prestigious institution recognized globally for its strong emphasis on applied research, innovation, and interdisciplinary collaboration. The university actively encourages projects that bring together engineering, computer science, and emerging digital technologies to address real-world problems.

ABOUT THE ORGANISATION

Woosong University is a well-known private university located in Daejeon, South Korea. It is recognized for its strong focus on practical education, research, and innovation. The university's mission is to prepare students for the modern, technology-driven world by providing hands-on learning experiences and encouraging creative problem-solving. Its vision is to become a global leader in education by combining academic excellence with real-world applications, particularly in fields like engineering, computer science, business, and hospitality. The organizational structure of the university includes several colleges and departments, each specializing in different areas of study, along with dedicated research centers that focus on advanced technologies and sustainable solutions. The Department of Electrical and Energy Engineering, in which I completed my internship, offers programs and research opportunities in power systems, renewable energy, automation, and smart technologies. Woosong University provides various services, including academic programs, research projects, industry collaborations, and modern laboratory facilities. During my internship, I mainly interacted with the Department of Electrical and Energy Engineering and its research labs, where I got the opportunity to work on AI- and IoT-based microgrid operations and learn from experienced professors and researchers.

INTERNSHIP ROLE AND RESPONSIBILITIES

I worked in the Department of Electrical and Energy Engineering at Woosong University, under the guidance of Dr. Salkuti Surender Reddy. My main responsibility was to explore and understand the concepts of AI- and IoT-based microgrid operations. As part of my daily and weekly tasks, I studied various research papers related to smart grid monitoring, forecasting, and optimization to build a strong theoretical understanding of the topic. Using this knowledge, I prepared a mini-project where I designed and simulated a smart grid monitoring and forecasting system. This system aimed to demonstrate how real-time data from the grid can be collected, analyzed, and used for better decision-making. Dr. Salkuti provided continuous guidance throughout the process, clarifying my doubts and helping me improve the design. For the simulation and design work, I used tools and technologies relevant to smart grid systems, ensuring that the project was both practical and aligned with real-world energy management applications.

Application of IoT in Microgrid Operation:

A microgrid is like a small power system that can produce, store, and supply electricity to a local area. It can work in two ways — either connected to the main electricity grid or completely on its own. A microgrid usually uses different sources of energy, such as solar panels, wind turbines, diesel generators, and batteries for storing power.

As our traditional power networks are turning into smart grids, there is a growing need for faster automation, real-time control, and smart decision-making. The problem is that normal control systems often face difficulties when dealing with the unpredictable nature of renewable energy sources and the complex mix of different power sources. This is where IoT and AI make a big difference. IoT helps collect real-time data from all the devices in the system, and AI uses this data to give smart, timely decisions that make the microgrid run more efficiently and reliably.

Advantages:

- Provides real time information
- Helps increasing productivity
- Enhances product quality
- It saves time and money
- Increase in the level of comfort and automation of the users
- Helps in monitoring the overall business processes
- Reduces waste and uses our natural resources effectively
- Predict and analyse the outcome of the operation
- Takes timely action to prevent the accident
- Minimise the human effort
- Identify the fault location
- Traffic or transportation tracking , inventory control , delivery, individual control tracking and customer management all can be made cost-efficient right tracking system.

Uses of IoT in general:

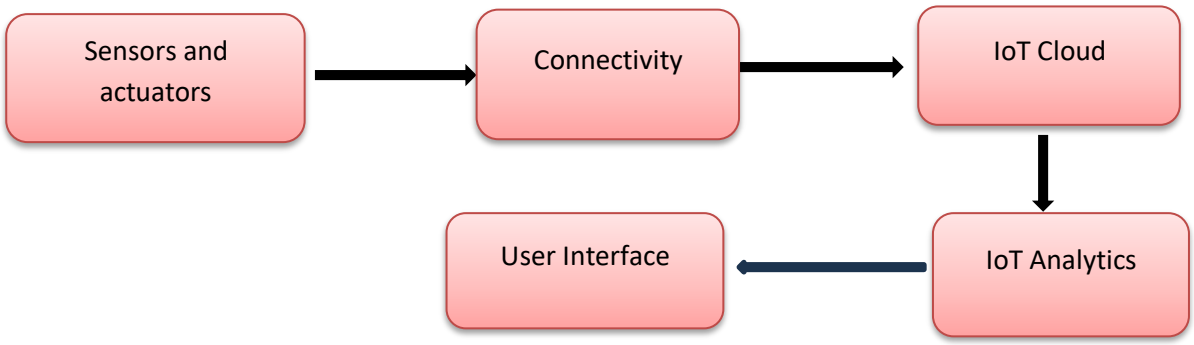
- Online Shopping- Instead of standing in queues and waiting for the salesperson to make us show their variety we can just quickly go grocery shopping for clothes and everyday used things online. Most people prefer it from a different e-commerce application like Amazon as it is way more reliable. It is both a time and money saver. IoT has set specific devices which connects online shopping through android and iOS phones.
- Health- IoT in healthcare has particular devices like beds in hospitals and unique types of sensors which are attached to our bodies as they can monitor our blood sugar continuously and

can give update about our current health checkup. These devices help measure blood pressure, and an oximeter regulates our body temperature. These devices are built with the current IoT systems. Thus, we can manage things at home instead of conitously visiting doctor and hospitals.

- In smart watches IoT provides real-time health and activity monitoring. These devices track steps, heart rate, sleep patterns, and even detect irregularities like abnormal heart rhythms. Through Bluetooth or Wi-Fi, smartwatches sync with smartphones to receive notifications, calls, and messages.
- Energy Saving- Different types of sensors are used in the different devices to save electric and water energy. It prevents the wastage of energy by making a connection between the communication and energy power user and supplier. These devices also help in monitoring energy consumption regularly.

Working of Internet of things:

Process followed in the functioning of internet of things comprises of multiple layer and stages such as data collection processing and communication in devices. In each layer of Internet of Things Its architecture provides us a proper framework for organizing the interconnected components like devices sensors , networks , cloud services and all applications in IoT surroundings. The working of IoT can be broadly categorized into five key stages: sensing, connectivity, data processing, cloud integration, and user interaction. Each of these layers of IoT plays a different role in making sure that the IoT system is operating properly, effectively, and efficiently. Sensors gather or collects all the data and convert it into actionable form or insights layers of IoT plays major role in it.



SENSORS AND ACTUATORS:

Sensors measure physical quantities from the surrounding and convert them into electrical signals analog or digital. Sensors are integrated with microprocessors or microcontrollers in their embedded systems. With the enhancement in technologies nowadays modern sensors are compact and efficient. The process of selection of a sensor depends on factors such as accuracy, precision, intelligence level, and the specific needs of the application.

Following are the types of sensors:

- Temperature sensor: It detects the heat and cold. Used in Smart thermostats, fridges.
- Motion sensor: It detects the movement. Used in Security systems, automatic lights.
- Proximity sensor: It detects how close is the object. Used in Parking sensors in cars.
- Light sensor: It detects the light level. Used in Auto-brightness in phones.
- Humidity sensor: It detects the moisture in the air. Used in Weather stations, greenhouses.
- Gas sensor: It detects the gas concentration in the air. Used in Smoke detectors, air quality monitors.
- Pressure sensor: It detects the pressure of the force. Used in Industrial machines, car tires.
- Sound sensor: It detects the sound or vibration. Used in Voice assistants, baby monitors.

Actuators are devices that receive signals ,usually electrical signals and perform physical actions such as moving, turning, opening, or adjusting objects. Actuators are very important as they carry out automated actions based on data from sensors. The control signals sent to actuators can be analog or digital, and they are often used to turn devices on or off.

Example: In a smart fan, a temperature sensor detects the room's heat. The system processes this data and sends a signal to the actuator, which then turns the fan on automatically.

Following are the types of actuators:

- Electrical actuators: It produces motion using electrical energy. Used in Turning on a light bulb, smart fans.
- Hydraulic actuators: It uses fluid pressure to create movement. Used in Heavy machinery and robotics arm.
- Pneumatic actuators: It uses compressed air to produce movement. Used in air brakes , automatic doors.
- Thermal actuators: It expands or contracts with heat. Used in thermostats, temperature based valves.
- Mechanical actuators: It converts energy into physical motion. Used in locks , lifts , motors.

CONNECTIVITY:

IoT connectivity is the means by which an IoT device, which can be anything from a simple sensor in a factory to a self-driving vehicle and has variety of applications from streetlights to robots, connects to the cloud, other devices and integration points such as IoT gateways. IoT connectivity is very crucial because without it devices are just things no effective work can be done without it. It is the IoT connectivity that brings value from IoT by communicating their data to enable action to be taken like different services are delivered and revenue is generated. A wide variety of wired and wireless protocols are used to meet up the variety of needs of IoT applications.

Some commonly used connectivity options based on range include:

- For short Range : It means for the object that are in the distance of less than 300ft. Bluetooth , Zigbee , Wi-Fi , NFC and Z-Wave are suitable for home automation, wearables, and small-scale applications.
- For medium Range: It means for the object that are in the distance between 300ft to 5 km. Cellular Networks like 2G, 3G, 4G, and 5G are widely used for mobile tracking, vehicle monitoring, and smart city solutions.
- For long Range: It means for the object that are in the distance between 5km to 100km. LoRaWAN, Sigfox , NB-IoT(Narrowband-IoT) are used in applications like smart agriculture, environmental monitoring, and industrial IoT where devices are spread across large areas.

Other emerging and specialized connectivity options include:

- **Ethernet** – For wired, high-speed, stable industrial or office connections.
- **Satellite IoT** – For remote locations like oceans, forests, or deserts where terrestrial networks are not available.
- **6LoWPAN** – Used in smart grids and remote monitoring systems for IPv6-based communication over low-power wireless networks.

The correct protocol ensures that the system is efficient, cost-effective, and suited to the specific use-case of the IoT deployment.

IoT CLOUD:

The IoT Cloud plays a central and critical role in the architecture of IoT systems. One component that improves the success of the Internet of Things is Cloud Computing. Cloud computing enables users to perform computing tasks using services provided over the Internet. The use of the Internet of Things in with the cloud technologies has become a major contributor to the digital technology. the Internet of Things and cloud computing are now related to each other. In short, the IoT Cloud

serves as the brain of the IoT ecosystem, transforming data into meaningful work and enabling real-time communication between devices, users, and systems.

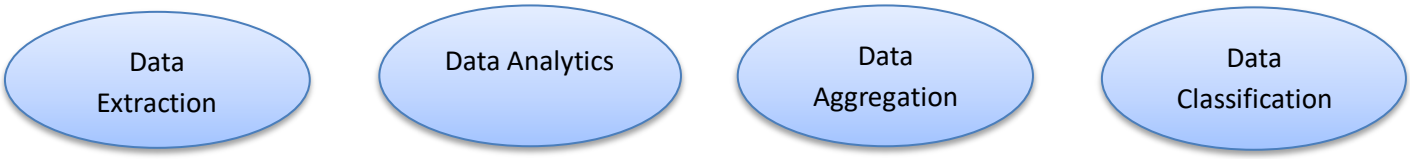
Benefits And Functions of IoT Cloud:

- IoT Cloud Computing provides many connectivity options, implying large network access. People use a wide range of devices to gain access to cloud computing resources: mobile devices, tablets, laptops. This is convenient for users but creates the problem of the need for network access points.
- Developers can use IoT cloud computing on-demand. In other words, it is a web service accessed without special permission or any help. The only requirement is Internet access.
- Based on the request, users can scale the service according to their needs. Fast and flexible means you can expand storage space, edit software settings, and work with the number of users. Due to this characteristic, it is possible to provide deep computing power and storage.
- Cloud Computing implies the pooling of resources. It influences increased collaboration and builds close connections between users.
- As the number of IoT devices and automation in use grows, security concerns emerge. Cloud solutions provide companies with reliable authentication and encryption protocols.

IoT ANAYLTICS AND DATA MANAGEMENT:

IoT devices constantly generate a large volume of raw data from various sources such as sensors, actuators, and embedded systems. Managing of the data should be done efficiently to convert it into actionable work. IoT data management includes the collection, storage, organization, security, analysis, and interpretation of this data in real time.

It consists of some key component:



Without proper data management, even the most advanced devices would not be able to deliver smart and optimized results. IoT analytics allows organizations to gather deeper knowledge, improve operational and thoughtful efficiency, and make better and impactful decisions across industries such as healthcare, agriculture, energy, and transportation.

USER INTERFACE:

The User Interface acts as the bridge between the user and the IoT system. Its agenda is to monitor and control the field devices by allowing users to interact with the data collected from sensors and the actions triggered through actuators. A well-designed UI makes the IoT system user-friendly and user-worthy, efficient, and accessible.

Various types of communication methods are used to notify the user, such as:

- **Notifications** on smartphones or desktops
- **Emails** for logs, updates, or alerts
- **Real-time alerts** for emergency conditions
- **Alarms** (audible or visual) for safety or operational warnings

Through the interface, users also sends commands to IoT devices. For example, adjusting thermostat settings, turning on lights, locking doors, curtain rolls or scheduling irrigation in a smart farm. Through user interface it works more efficiently and ensures full control over remote. Provides more user-friendly experience.

Moreover, the User Interface (UI) may display various types of information to enhance user experience and system control. It can provide live video and real-time sensor readings which allows us to monitor the ongoing actionable insights. Interactive dashboards with data charts and analytics help in visualizing the system's current status and performance also helps user to keep an eye. Additionally, the UI often shows previous data which helps user to track their data and analyse their use and efficiency. Some advanced interfaces also provides predictive suggestions, which are generated using AI or machine learning algorithms, to help users make informed decisions and anticipate future outcomes.

WORKING OF IOT IN MICROGRID:

1. Real-Time Monitoring

The smart grid monitoring system uses **IoT-based sensors** installed at different points in the microgrid, as shown in the images. These sensors continuously collect operational data, enabling precise supervision and early detection of faults.

- **Solar PV Inverters** – Measure voltage, current, and solar irradiance to assess generation efficiency. The data helps in tracking daily energy production, identifying shading or panel faults, and optimizing inverter performance.

- **Wind Turbine Controllers** – Monitor wind speed, rotor speed, generated power, and mechanical stresses. This ensures turbines operate within safe limits and respond efficiently to varying wind conditions.
- **Battery Management Systems (BMS)** – Keep track of state of charge (SOC), state of health (SOH), temperature, and voltage. Accurate battery data prevents overcharging or deep discharge, extending battery life and ensuring stable energy storage.
- **Load Points / Smart Meters** – Measure real-time power consumption, frequency, power factor, and phase imbalance at various locations. This information helps balance loads, reduce losses, and improve power quality.
  The collected data is transmitted via reliable **communication protocols** like *Modbus*, *Zigbee*, *Wi-Fi*, *LoRaWAN*, or *Ethernet* to a Central Control Unit (CCU) or Supervisory Control and Data Acquisition (SCADA) system. Here, operators can visualize the system's status through dashboards, perform trend analysis, and make informed operational decisions. Advanced analytics can also forecast generation and demand patterns, improving grid stability and reducing outages.

2. Remote Control and Automation

Remote control and automation turn a microgrid from a manually-operated system into a self-managing network. IoT devices collect live measurements, while controllers (local and central) send setpoints to devices like breakers, inverters and relays. Automation reduces response time, improves reliability, and enables fast fault isolation, load management and energy optimisation — often without human intervention.

Key components involved

- **Intelligent Electronic Devices (IEDs)** / protection relays — perform fast protective actions and can be commanded remotely.
- **Circuit breakers, contactors and motorised switches** — actuated remotely to connect/disconnect feeders or loads.
- **Inverters / converters** — accept setpoints for active/reactive power, operate in grid-following or grid-forming modes, control ramp rates and execute MPPT for PV.
- **Battery Management System (BMS) & Battery Inverter** — manage charging/discharging, SOC limits and thermal protections.
- **PLCs / RTUs / Edge controllers** — implement local automation logic and interface between field devices and the Central Control Unit (CCU).
- **Central Control Unit (CCU) / Microgrid Central Controller (MGCC) / DERMS** — perform higher-level optimisation, scheduling and coordination.

- **HMI / SCADA** — human interfaces for monitoring, alarms, manual overrides and logging.

Communication Protocols & Links

Effective microgrid automation relies on robust communication channels between field devices, units, and cloud platforms. **Field protocols** such as MODBUS , DBP3, IEC 61850 and OPC UA control ensure reliable data exchange between Intelligent Electronic Devices (IEDs), sensors, and substation equipment. For higher-level data handling and integration with remote dashboards, **IoT/cloud protocols** like MQTT , HTTP/REST and AMQP are used to transmit telemetry, alerts, and control commands.

Both **wireless and wired media**—including Ethernet, Wi-Fi, 4G/5G, LoRaWAN, and RS-485—provide connectivity, often with redundant communication paths to ensure uninterrupted operation. Time-synchronization technologies such as NTP or GPS clocks maintain coordinated control actions across the system.

Automation is embedded into this communication framework, enabling the microgrid to respond instantly to changing conditions. For instance, sensor data can trigger automated inverter adjustments, load shedding, or fault isolation without waiting for human intervention. Communication protocols ensure these automated actions are delivered accurately and in real time, allowing the system to function seamlessly and efficiently.

3. Energy Data Analytics

In a smart microgrid, IoT systems collect information from devices like meters, sensors, and controllers. This information is stored in cloud databases, where both current data and past records are available. By studying this data, operators can understand how much power is being produced and used, and how this changes during the day, week, or season.

With these insights, it becomes easier to predict when electricity use will be highest and prepare in advance. For example, more generators or storage systems can be used during peak hours, and less important loads can be run when demand is low. Data analytics can also warn about possible equipment problems—if a machine starts showing unusual patterns, it can be fixed before it fails.

This process also helps decide which power source to use at the right time—whether it’s solar, wind, batteries, or backup generators—so energy is used in the most efficient and cost-effective way. Over time, the system “learns” from past data and can make quicker, smarter decisions during emergencies like bad weather or a sudden loss of power supply.

Role of AI in Microgrid Operation:

Artificial Intelligence enhances microgrid operations through a wide range of intelligent functionalities. Its primary contributions include:

1. Energy Forecasting

Energy forecasting means predicting how much electricity will be needed and how much will be produced in the future. AI tools like Support Vector Machines (SVMs) and Long Short-Term Memory (LSTM) networks study past energy use, weather reports, and seasonal patterns to make these predictions. For example, if the forecast says there will be cloudy days, AI knows solar panels will produce less power and can prepare other sources in advance. This helps operators decide when to make more energy or store it in batteries so there is no power shortage. Good forecasting also means less use of fuel-based backup generators, which saves money and reduces pollution. For instance, in a coastal microgrid, AI predicted strong winds for two days and charged the batteries in advance while reducing generator use, which cut costs and saved fuel.

2. Energy Management

AI helps decide how and when to use or store electricity in a microgrid. It balances the power supply between buildings, renewable sources like solar and wind, and batteries. Using both live and past data, AI can choose the best time to charge batteries or use stored energy. This avoids wasting power and keeps the supply steady even during busy times. For example, if the batteries are already full and demand is low, AI will store extra renewable power for later instead of letting it go unused. This not only saves energy but also helps batteries last longer and supports cleaner power use.

3. Fault Detection and Predictive Maintenance

This is about finding and fixing problems in the microgrid before they cause a breakdown. Special IoT sensors are placed in equipment like batteries, solar inverters, and generators. These sensors check things like temperature, noise, and vibrations. If AI notices something unusual—like a machine getting too hot or making strange noises—it sends an alert. This way, the maintenance team can fix it before it fails completely. This approach is called predictive maintenance and helps avoid costly repairs and long power cuts. For example, AI once noticed that a solar inverter was getting hotter than normal. The team checked and found the cooling fan was failing. They replaced it before the inverter stopped working, avoiding a two-day blackout.

4. Grid Optimization

Grid optimization means making sure all parts of the microgrid work together in the best way possible. AI acts like the brain of the system, deciding which energy source to use first—usually renewable sources like solar because they are free and clean. If extra power is available, it stores it in batteries. If more power is needed, it takes it from storage or, as a last resort, uses generators. AI also looks at electricity prices in real time. If buying electricity from the main grid is cheap, it may use that instead of running generators. The goal is to make the microgrid reliable, affordable, and eco-friendly. For example, on a sunny day, AI used only solar power and stored the extra in batteries. Later, during the evening peak, it used the stored energy before switching to generators, which reduced fuel usage.

5. Demand Response Management

Demand response management is about handling times when electricity usage suddenly increases a lot. High demand usually happens in the summer when many people use air conditioners at the same time. This can put stress on the grid or even cause blackouts. AI helps by finding non-essential energy use and turning it off or delaying it until the demand goes down. It can also suggest that people use certain appliances at night or early morning when electricity is cheaper and more available. This not only keeps the grid stable but also saves money for both the microgrid operators and the users. For example, on a hot afternoon, AI might temporarily switch off water heaters or delay charging electric cars to keep the grid from overloading.

Advancements in AI-Driven Microgrids: Some of the latest and most exciting advancements in the use of AI in microgrids include:

- **Reinforcement Learning (RL):**
  Reinforcement Learning is a special kind of AI that learns by doing and improving over time. Just like a human learns from trial and error, RL allows a microgrid system to make decisions, see the results, and learn what works best. For example, it can learn when to store energy or when to buy electricity from the grid based on changing weather, electricity prices, or user demand. The more it learns, the smarter it becomes at saving energy and money while keeping the power supply stable.
- **Hybrid AI Models:**
  Researchers are now combining different types of AI methods, such as fuzzy logic, genetic algorithms, and neural networks, to handle complex problems in microgrids. Each method has its strengths. For instance, fuzzy logic deals well with uncertainty, genetic algorithms find the best possible solutions by mimicking evolution, and neural networks are great at learning patterns. When these are used together, the hybrid model becomes more powerful and accurate, especially in real-time energy management and system optimization.

- **Working with IoT and Edge Computing:**
  AI is now being integrated with Internet of Things (IoT) devices, which collect real-time data from sensors placed on solar panels, batteries, and other equipment. With edge computing, this data is processed close to where it is collected—on local devices instead of being sent far away to the cloud. This makes decision-making much faster, which is important for quick actions like switching power sources or fixing a fault. The result is a more responsive and reliable microgrid.
- **Blockchain with AI:**
  Blockchain is a secure way of recording information, and when it's combined with AI, it can improve trust and transparency in how microgrids operate. In a local energy system where people might sell or share electricity with each other, blockchain can keep secure records of who produced how much energy, who used it, and how much money is owed. AI can analyze this data to automate billing, detect fraud, and ensure fair pricing. Together, blockchain and AI help make microgrids smarter, safer, and more efficient.

PROJECT REPORT

1. INTRODUCTION

The increasing complexity of modern power grids demands intelligent monitoring and forecasting systems to optimize electricity distribution, improve grid stability, and reduce energy losses. This project aims to design and simulate a **Smart Grid Monitoring and Forecasting System** that leverages virtual IoT data, machine learning models, and an interactive web-based dashboard. The system intelligently monitors electrical parameters, predicts future load demand, and presents actionable insights for grid management.

2. OBJECTIVES

- To implement machine learning techniques to forecast load demand based on historical data.
- To simulate real-time smart grid data including voltage, current, frequency, and load using virtual IoT sensors.
- To develop a user-friendly web dashboard to visualize live data trends and predicted load, enabling better grid management decisions.
- To optimize electricity distribution and anticipate peak loads, thereby improving grid efficiency and reliability.

3. TOOLS AND TECHNOLOGY USED

In this section, it is explained the results of research and at the same time is given the comprehensive discussion. Results can be presented in figures, graphs, tables and others that make the reader understand easily.

- **Python**: For data simulation, processing, and machine learning (libraries like pandas, scikit-learn, matplotlib).
- **Node.js and Express**: Backend server to serve data and host the web application.
- **CSV**: Data format used for storing simulated smart grid data.
- **Git and GitHub**: Version control and remote repository hosting.

4. METHODOLOGY

a. Data Simulation

Virtual IoT data was generated using a Python script that simulates 100 data points of key electrical parameters:

- Voltage (210-240 V)
- Current (5-20 A)
- Frequency (49-51 Hz)
- Load (calculated from voltage and current)

The data was saved into a CSV file (smart_grid_data.csv) mimicking real-time sensor output with timestamps.

4.2 Load Forecasting Model

Using the historical simulated data, a machine learning regression model was built in load_predictor.py to predict future load values. The model employs algorithms such as Linear Regression to forecast based on voltage, current, and frequency trends.

4.3 Backend Server and API

A Node.js server was created (app.js) using Express to serve the CSV data as JSON through an API endpoint (/data). This allows the front-end to fetch live data dynamically.

4.4 Web Dashboard

The dashboard is a responsive web application designed to provide an intuitive and interactive user experience. Users can seamlessly switch between chart types and refresh the data through built-in controls, enhancing usability. The application fetches data dynamically from the backend server via an API, ensuring real-time updates. Additionally, the interface is styled using CSS to maintain a clean, modern, and user-friendly appearance.

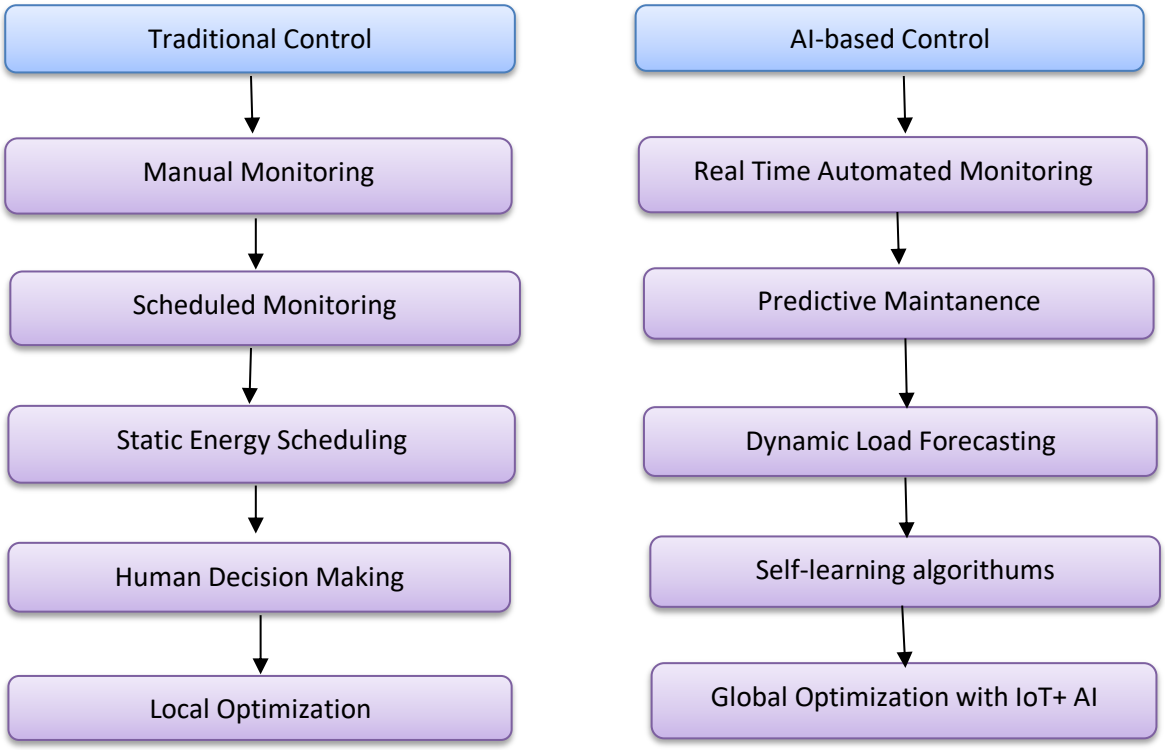
5. RESULTS

The project successfully generated virtual smart grid data that accurately simulates real electrical parameters with realistic variations, such as voltage, current, frequency, and load. A regression-based machine learning model was developed to predict future load values with reasonable accuracy using historical data inputs. Furthermore, a responsive and interactive web dashboard was built to visualize both actual and predicted load values, enabling effective real-time monitoring and forecasting. Overall, the system equips grid operators with valuable insights to anticipate peak demand and make informed decisions for efficient electricity distribution.

FUTURE SCOPE:

The application of Artificial Intelligence in microgrid operation is still evolving, and the future holds vast potential for innovation. Some of the key developments and trends expected to shape the future include:

- **Integration with Smart Cities**: In the future, microgrids will not operate in isolation but will become interconnected with smart homes, electric vehicles, intelligent traffic systems, and digital public services. AI will play a central role in ensuring these systems communicate efficiently and share energy resources dynamically. For instance, during periods of low energy demand, electric vehicles can serve as energy storage units, feeding power back into the grid.
- **AI-powered Disaster Response:** During emergencies such as floods, earthquakes, or cyclones, these systems can continue functioning independently of the main grid. AI will ensure that available power is prioritized for essential services like hospitals, water treatment plants, and emergency shelters. It can also analyze sensor and weather data to anticipate disaster impacts and proactively reconfigure the grid to minimize damage. By ensuring uninterrupted power to vital systems, AI-powered microgrids can save lives and speed up disaster recovery efforts.
- **Improved Energy Access in Remote:** In regions where traditional power grids are not feasible, AI can help design, operate, and maintain cost-effective microgrids. It can analyze local energy consumption patterns, weather data, and resource availability to optimize the configuration of energy sources like solar, wind, and battery storage. Additionally, AI can predict maintenance needs and troubleshoot problems remotely, reducing the need for constant on-site human intervention.



DRAWBACKS AND CHALLENGES: Even though AI brings many advantagess to microgrids, there are still many challenges to for us to overcome it. One of the major issue is the high initial cost, as setting up an AI-based microgrid requires advanced devices, software systems, skilled workers, and strong network infrastructure. Another concern is cybersecurity, because everything is connected, and this increases the risk of hacking or data theft, which could disrupt the system or leaks sensitive information. Also, data quality is very important—AI needs accurate and reliable data to make smart decisions, but if the data is incorrect or incomplete, it can lead to wrong actions. Additionally, managing and maintaining AI systems requires skilled professionals, and a lack of trained experts can make it difficult to operate such systems properly. Lastly, differences in rules and standards across regions and companies can make it harder to apply AI technologies everywhere in a smooth and unified way.

CONCLUSION

Artificial Intelligence (AI) is changing the way microgrids work by making energy use smarter and more efficient. AI helps predict how much energy will be needed and when, so power is available without wasting resources. It also manages energy storage and use, making sure batteries and backup systems work at the right time.AI can detect problems early by monitoring equipment, helping to avoid big failures and costly repairs. It controls the microgrid to use clean energy first and only use backup power when needed, which saves money and helps the environment. AI also helps manage high power demands by reducing or shifting usage to prevent outages. In short, AI makes microgrids more reliable, cost-effective, and eco-friendly. As more renewable energy is used, AI will be important to keep our energy systems smart and sustainable in the future.